



جامعة الملك عبد الله
للعلوم والتقنية
King Abdullah University of
Science and Technology

Technology Innovation and Entrepreneurship (TIE)

Transforming Ideas into
Innovations

Feb 23, 2026



Accelerating Impact: A New Strategy for KAUST

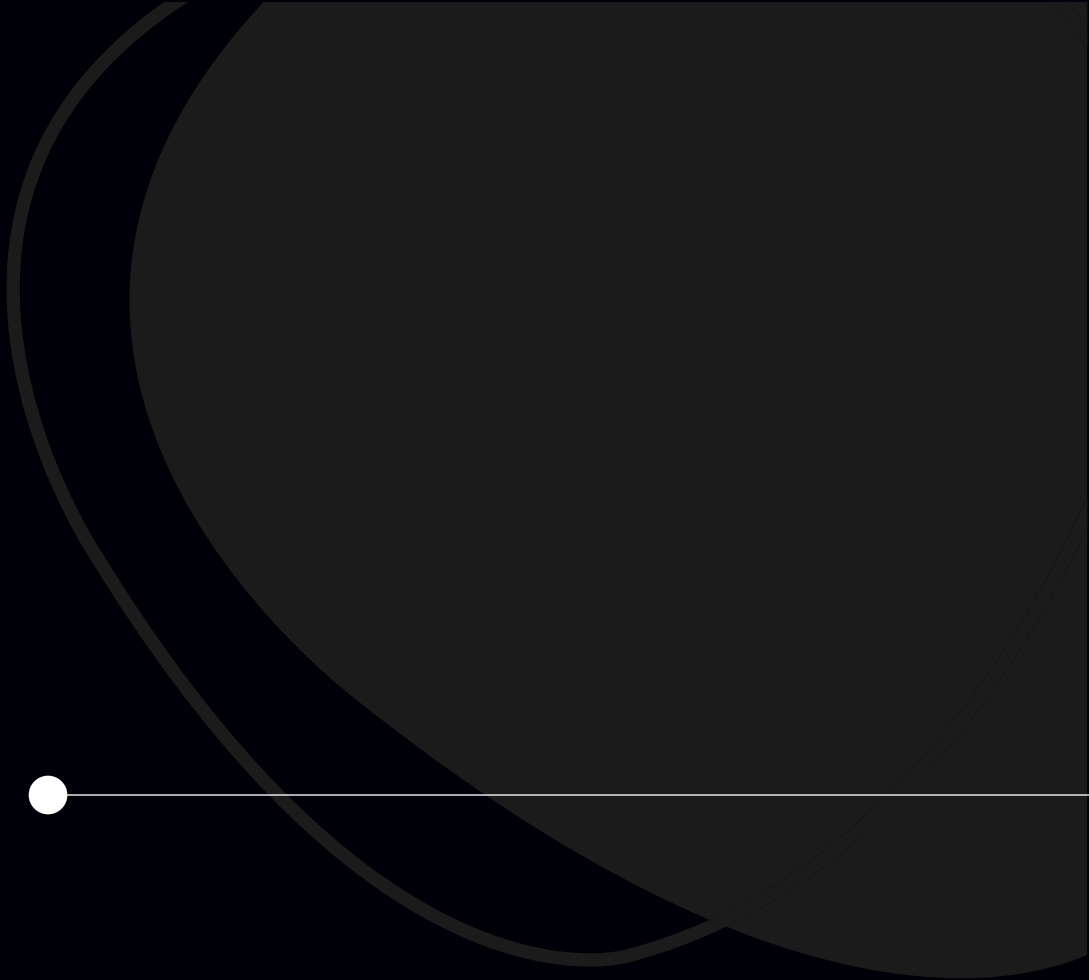
KAUST's new strategy presents new opportunities to accelerate the University's positive impact on the world through alignment with the Kingdom's Vision 2030 Research, Development, and Innovation (RDI) priorities, enhancing our global and local partnerships and **focusing on commercialization and economic impact.**



“ [KAUST] has ... become one of the leading research universities in the world. The new strategy ... represents a new era for the University to become a beacon of knowledge and a source of inspiration and innovation ... for the betterment of the Kingdom and the world. ”

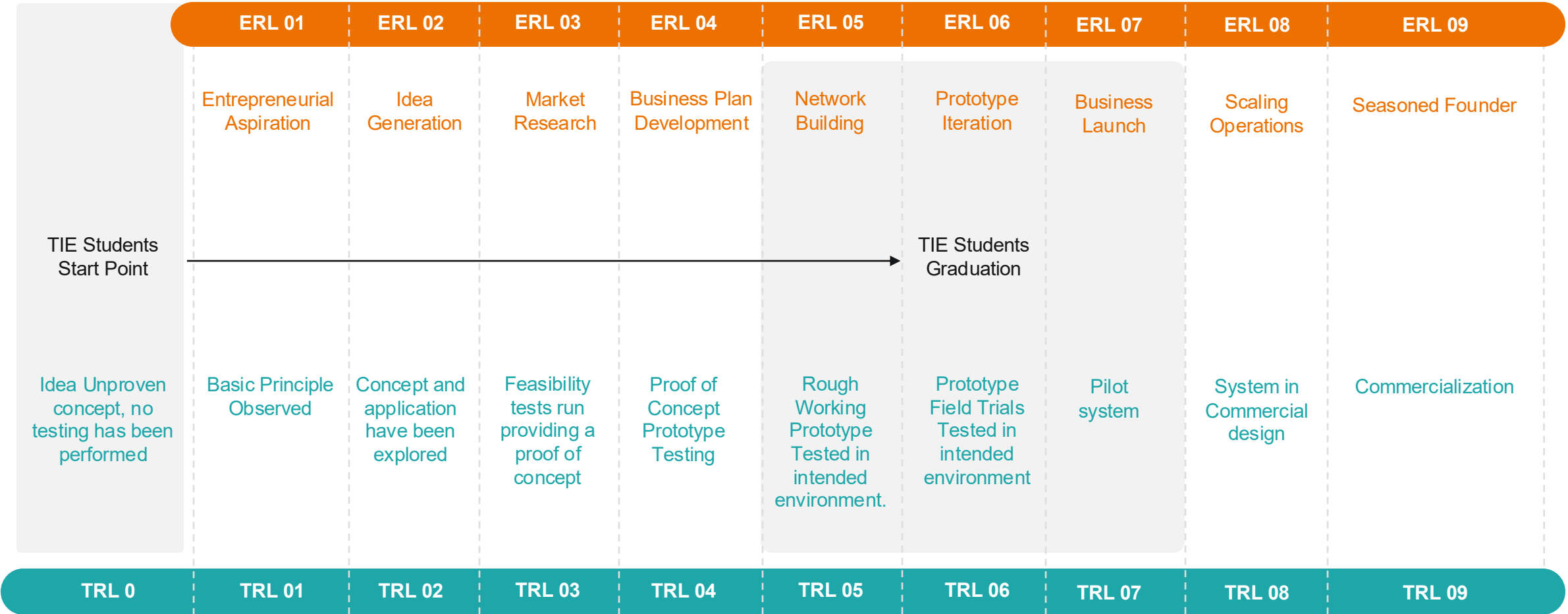
– *Chairman of KAUST Board of Trustees Crown Prince Mohammed bin Salman August 2023*

TIE Program Vision



- Curating world-class **technology founders and Innovation leaders** in Saudi Arabia and beyond, **driving innovation and economic diversification** through cutting-edge **entrepreneurship education and training**

ENTREPRENEURSHIP READINESS LEVEL (ERL)



TECHNOLOGY READINESS LEVEL (TRL)

TIE Master of Science (M.Sc.)

Graduates of TIE Develop Six Core Competencies

01 Team Leadership & Collaboration

Build strong, multidisciplinary teams and learn how to lead, communicate, and collaborate effectively under uncertainty.

02 Problem Discovery

Identify real, high-impact problems by applying design thinking methodologies.

Develop the ability to frame the right questions before jumping into solutions.

03 Technology-to-Impact

Gain the ability to bridge science, engineering, and societal or industry needs.

04 Venture Design & Entrepreneurship

Develop the skills to design, test, and shape new ventures using modern entrepreneurship and business model frameworks.

05 MVP Development

Turn ideas into tangible solutions through rapid prototyping, experimentation, and hands-on MVP building.

06 Execution, Validation & Go-to-Market

Gain experience in market testing, pitching, and turning projects into real-world impact.

TIE Program

On-Campus Activities

1

Technology Innovation
and Entrepreneurship
(TIE) **MSc degree**

2

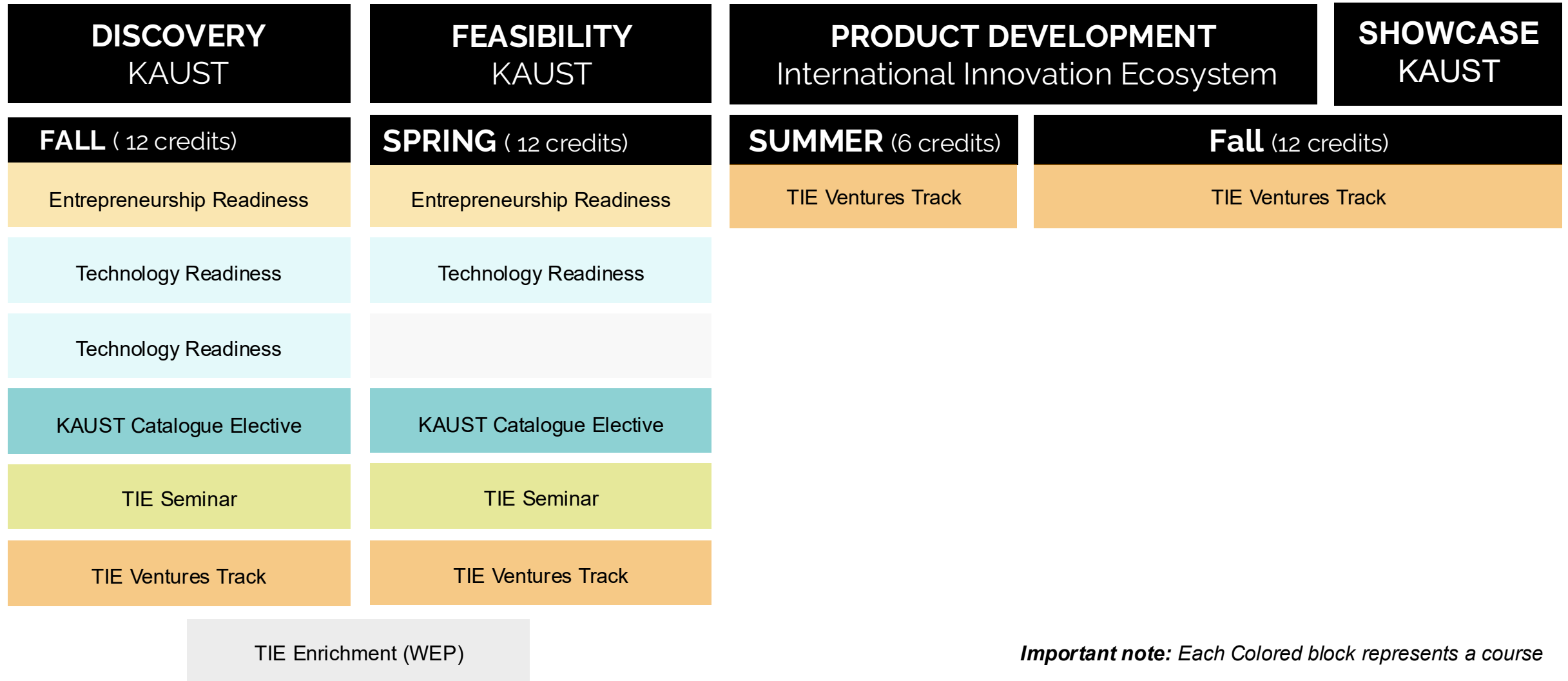
**Open Electives for
all KAUST Students**

3

**Innovation training
and Certification**

TIE Master of Science (M.Sc.)

Students' Journey



Important note: Each Colored block represents a course

TIE Master of Science (M.Sc.)

Entrepreneurship Readiness Courses



MIKKEL DRAEBYE

Instructional Professor
Entrepreneurship

DISCOVERY Spring Semester

TIE 211 From Idea to Viable Business Concept

- 01. Innovation management
- 02. Customer discovery
- 03. Market analysis
- 04. Value proposition
- 05. Business model design
- 06. Evidence-based entrepreneurship
- 07. Go-to-market strategy

FEASIBILITY Fall Semester

TIE 212 From Concept to Early-Stage Startup

- 01. Risk & uncertainty
- 02. MVP validation
- 03. Business model testing
- 04. Market & financials
- 05. Business planning
- 06. Pitching & storytelling
- 07. Corporate entrepreneurship

TIE Master of Science (M.Sc.)

Technology Readiness Courses



CARLOS GRANDE

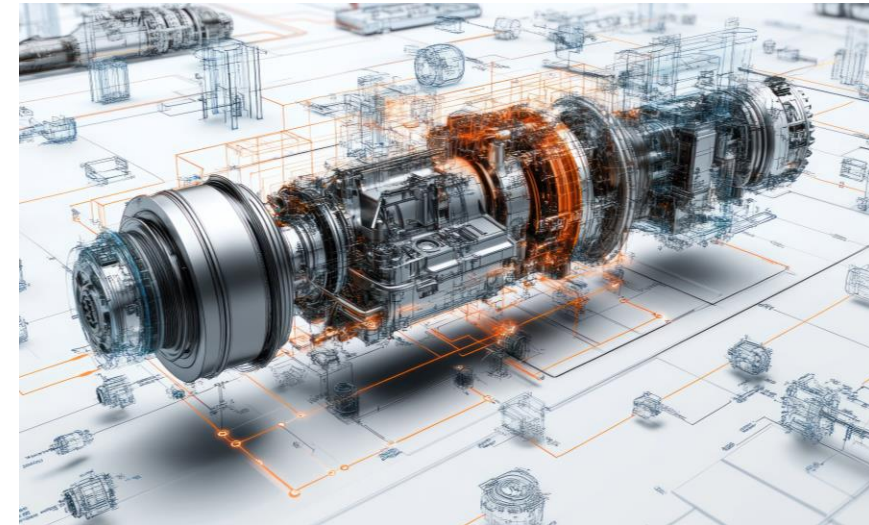
Associate Professor,
Chemical Engineering

DISCOVERY Spring Semester

TIE 241

Product Design and Digital Manufacturing

- 01. Design thinking for product innovation
- 02. Product design strategies & market fit
- 03. CAD and parametric modeling
- 04. Digital manufacturing & 3D printing
- 05. Prototyping & iterative design
- 06. Design for manufacturing/sustainability
- 07. Product branding and communication



TIE Master of Science (M.Sc.)

Technology Readiness Courses



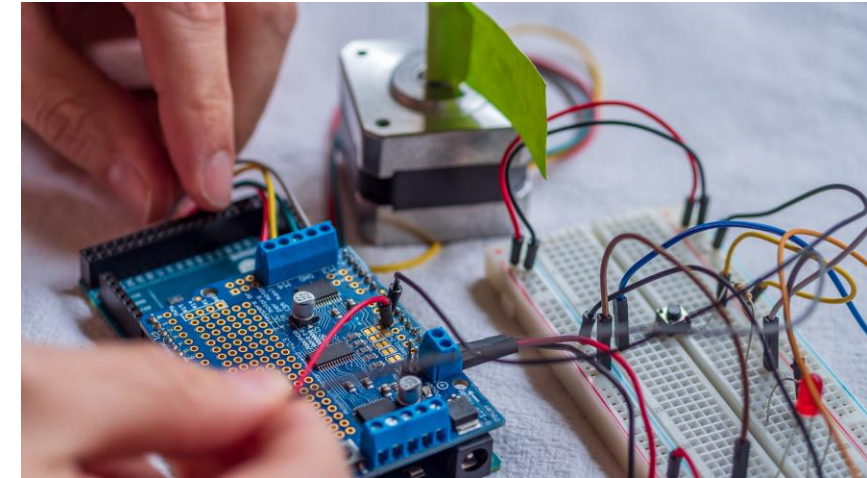
AMR ABDEL-HAMID

Instructional Professor
Engineering

DISCOVERY Spring Semester

TIE 203 Systems Design Studio

- 01. System-level problem solving
- 02. Embedded systems
- 03. Sensor integration
- 04. IoT connectivity, AIoT development
- 05. Modular system architecture
- 06. Interface design
- 07. System integration, testing & iteration



TIE Master of Science (M.Sc.)

TIE Ventures Track

DISCOVERY	FEASIBILITY	PRODUCT DEVELOPMENT	SHOWCASE
01. Design Thinking 02. Team formation 03. Problem definition 04. Customer discovery 05. Problem validation 06. Value proposition 07. Competitive analysis	01. Ideation 02. Solution development 03. Business model testing 04. Specifications 05. Prototype iterations 06. User feedback loop 07. Branding development	01. Prototype iterations 02. MVP Development 03. Business model testing 04. Funding strategy 05. Traction and agreements 06. Innovation disclosure 07. Video Adv 08. Pitch deck finalization	01. Booth experience 02. Pitch 03. Launch
Outcome: • Venture approval	Outcome: • Working Prototype • Refined Business model • Execution plan	Outcome: • Early Traction • Showcase readiness • Working MPV	Outcome: • Start-up Launch • Or Job placement

TIE Student Profile

What we look for in candidates



1. Clarity of Purpose

Why TIE, why now?

- 90-second video sharing your motivation and vision
- Show, don't just tell (prototypes, ideas, ambition)



2. Entrepreneurial Exposure

Have you explored building before?

- Experience with business or entrepreneurship concepts
- Participation in competitions, initiatives, or ventures



3. Builder Mindset

Can you turn ideas into reality?

- Show hands-on experience in product development, prototyping, or design
- Clear understanding of your contributions



4. Vision & Ambition

Where are you headed?

- A defined direction as an innovator or entrepreneur
- How TIE fits into your 5-year trajectory



5. Ideas & Initiative

What problems are you drawn to solve?

- A business idea or challenge you're passionate about
- Evidence of curiosity, ownership, and drive

TIE Program

Master of Science (M.Sc.)

Innovation Ecosystem



TIE Program

Master of Science (M.Sc.)

Cohort 1, Ventures



TIE Master of Science (M.Sc.)

TIE Ventures Track



TIE Master of Science (M.Sc.)

Seminars



Type of Speakers:

1. Founders
2. Industry leaders
3. Venture capitalists
4. Corporate innovation
5. Scale-up operators
6. Policymakers
7. Ecosystem builders
8. Technology transfer experts

TIE Program At KAUST



TIE Program

National and International Partnership

هيئة تنمية البحث
والتطوير والابتكار
Research Development
and Innovation Authority

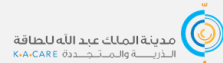


البرنامج الوطني
لتنمية تقنية المعلومات
National Technology
Development Program



National Industrial
Development and
Logistics Program

TIE Strategic Partners



HALLIBURTON



الهيئة الملكية لمحافظة العلا
Royal Commission for AlUla



KAUST Industry Collaboration Program (KICP)



TIE Education Partners

Burson
LANDOR

TIE Brand Education and
Communications Partners

TIE Mentors

TIE Investors

TIE Master of Science (M.Sc.)

Interest and applications

Cohort 1

1540

Interested
Candidates

343

Total
Applications

30

Admitted
Students

24

Cohort-1
Spring, 2025

Cohort 2

1798

Interested
Candidates

479

Total
Applications

39

Admitted
Students

34

Cohort-2
Fall 2025/26

TIE Program

Master of Science (M.Sc.)

Cohort 2. Bachelor Degree Awarding Universities

The Americas



Middle East and Asia



Europe



TIE Team



AAMIR FAROOQ

TIE Program Director
Professor, Mechanical
Engineering



MIKKEL DRAEBYE

Instructional Professor
Entrepreneurship



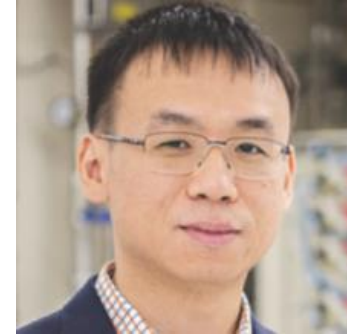
AMR ABDEL-HAMID

Instructional Professor
Engineering



KHALID NABIL SALAMA

Professor, Electrical and
Computer Engineering



XIAOHANG LI

Associate Professor, Electrical
and Computer Engineering



CARLOS GRANDE

Associate Professor,
Chemical Engineering



REHAM ALMUBARAK

Education and
Outreach Advisor



JOSE ILTON FILHO

Prototyping and
Lab Specialist



MARYAM MOHAMMED

Liaison and admin
assistant



WEDAD MUDAYFIN

Entrepreneurship
Program Coordinator
and Trainer
